DAILY PROJECT WORK LOG

Artery Segmentation and Stenosis Detection using Advanced Deep Learning Techniques

Objective

To develop an automated deep learning framework for accurate coronary artery lumen segmentation, centerline extraction, and quantitative stenosis detection and severity grading from angiographic images.

Workflow

- **Step 1: Data Preprocessing:** Input image normalization, contrast enhancement (CLAHE), and noise suppression.
- **Step 2: Artery Segmentation:** Deep learning model prediction of arterial lumen masks.
- **Step 3: Centerline & Geometry Profiling:** Skeletonization and orthogonal diameter measurement across the vessel path.
- **Step 4: Stenosis Quantification:** Local minimal lumen diameter calculation and percentage narrowing determination.
- **Step 5: Visualization & Output:** Visual overlay of segmented vessels and quantified stenosis findings.

Architecture

- **Model Architecture:** Attention U-Net / U-Net encoder-decoder network with multi-scale convolution and compound Focal + Soft Dice loss.
- **System Platform:** FastAPI backend, React + TypeScript UI, PostgreSQL database, secure file storage, and dedicated AI inference pipelines for X-ray and CT/CTA.

About It (Daily Work Done)

October 8, 2026

- **Define Project Objectives & Workflow:** Formalized core objectives for lumen segmentation, centerline extraction, and stenosis severity quantification; established 5-step sequential workflow.
- **Architecture Selection:** Specified Attention U-Net network backbone and compound Focal + Soft Dice loss formulation.
- **Documentation Setup:** Established project documentation structure and automated black-and-white PDF reporting system.

October 9, 2026

- **Select Medical Imaging Modality:** Selected 2D X-ray Coronary Angiography (XCA) as primary modality and 3D CCTA as complementary modality.
- **Select & Download Appropriate Dataset:** Acquired public/de-identified benchmark datasets (DCA1, ARCADE, ASOCA) for vascular segmentation and stenosis evaluation.
- **Understand Dataset Structure & Annotations:** Analyzed directory structure, image resolutions, binary lumen masks, and clinical stenosis bounding box annotations.

October 12, 2026

- **Design System Architecture:** Designed full-stack system architecture comprising React + TypeScript UI, FastAPI REST API, PostgreSQL database, secure storage, and decoupled AI services.
- **Design Database Schema:** Specified 14 relational tables covering users, patients, cases, images, preprocessing, analyses, findings, models, and reports.

- **Create UI Wireframes:** Designed wireframes for Login, Main Dashboard, Case & Upload Screen, Analysis Workspace, and Role-Based Access Control (RBAC).

October 15, 2026 (Today's Work)

- **Perform Exploratory Data Analysis:** Conducted EDA on angiographic images: evaluated intensity distribution, contrast variability, vascular pixel sparsity (<6%), and catheter/motion artifacts.
- **Set Up Frontend and Backend:** Initialized full-stack project scaffolds with React, TypeScript, and Vite on frontend, and FastAPI with modular API routing on backend.
- **Implement Authentication:** Built user authentication module with password hashing, JWT session token generation, session expiration, and role verification.
- **Implement Case Management:** Built case management service handling case creation, patient code linkage, modality tagging, status progression, and team access permissions.
- **Implement Image Upload:** Implemented multi-format file upload endpoint supporting DICOM, PNG, JPG, and NIfTI with file integrity verification and secure directory storage.
- **Build Initial Preprocessing Pipeline:** Constructed initial image preprocessing pipeline implementing intensity normalization, CLAHE contrast enhancement, and noise filtering.